MASTER 1 CYBERSECURITY (S1-ISI5-SÉCURITÉ)

Advanced Targeted Cyber Attacks

Vectors, Mechanics, and the AI Evolution

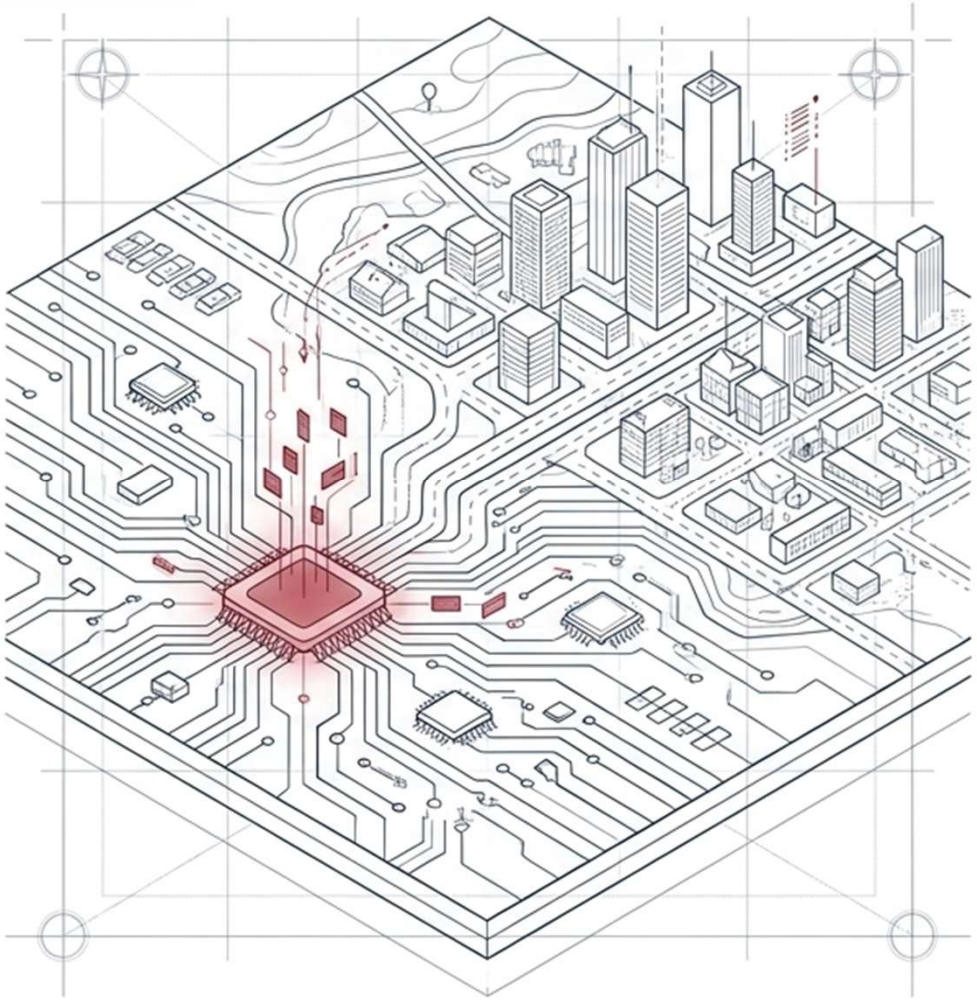
Deconstructing the anatomy of modern threats: from automated delivery to autonomous agents.

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UNIVERSITÉ PARIS 1 PANTHÉON-SORBONNE

Information Systems Security: Anatomy of an Evolving Battlefield

From foundational architectures to the industrialization of cybercrime and AI-driven risks in 2026.

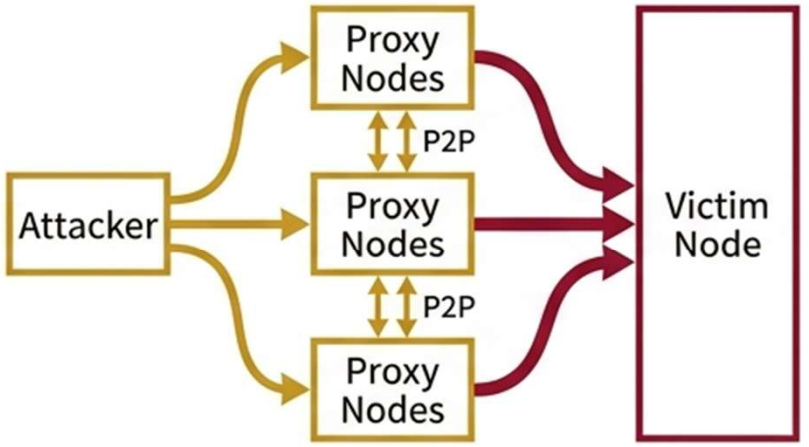


Yesterday
Direct & Rudimentary



- Direct IP access
- Unencrypted communications
- Basic infrastructure
- Single-actor execution

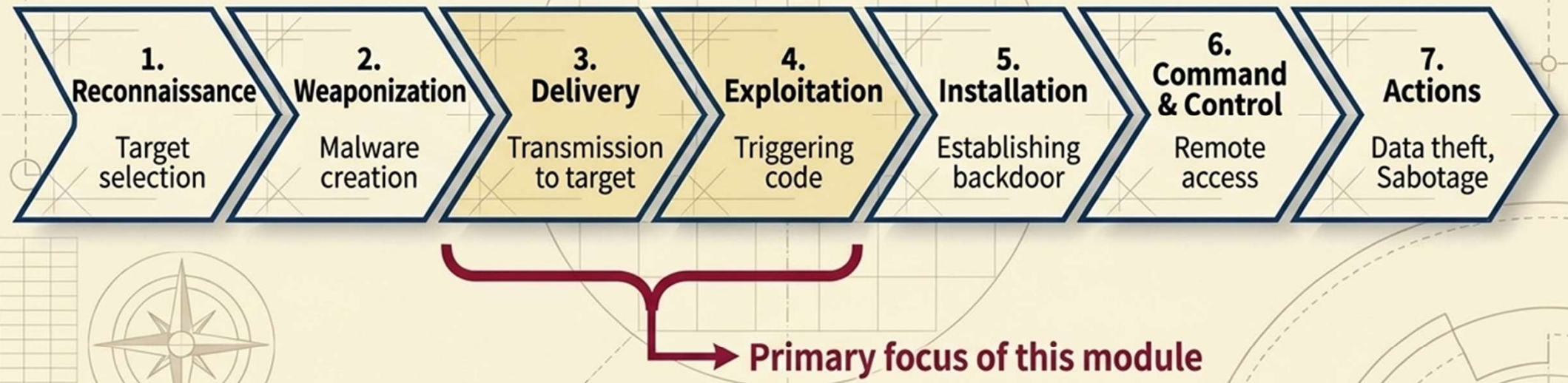
Today
Cascading & Resilient



- Anonymization via proxy cascades
- Redundant C2 nodes
- Heavy over-encryption
- Specialized technical actors (RaaS)

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The Universal Framework: The Cyber Kill Chain



Interrupting the chain at any single step successfully breaks the entire attack.

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Delivery Phase Deep Dive: Targeted Phishing & Spoofing

Spoofing: Falsifying Identity



- Modifies email headers and IPs to impersonate trusted entities
- Bypasses weak or missing SPF, DKIM, and DMARC protocols
- Goal: Borrow legitimacy

Spear-Phishing: Targeted Delivery



- Highly customized social engineering utilizing OSINT
- Delivers malicious attachments or links directly to specific users
- Goal: Extract credentials or execute initial payload

Example: The 2013 Target breach began with spear-phishing an HVAC vendor.

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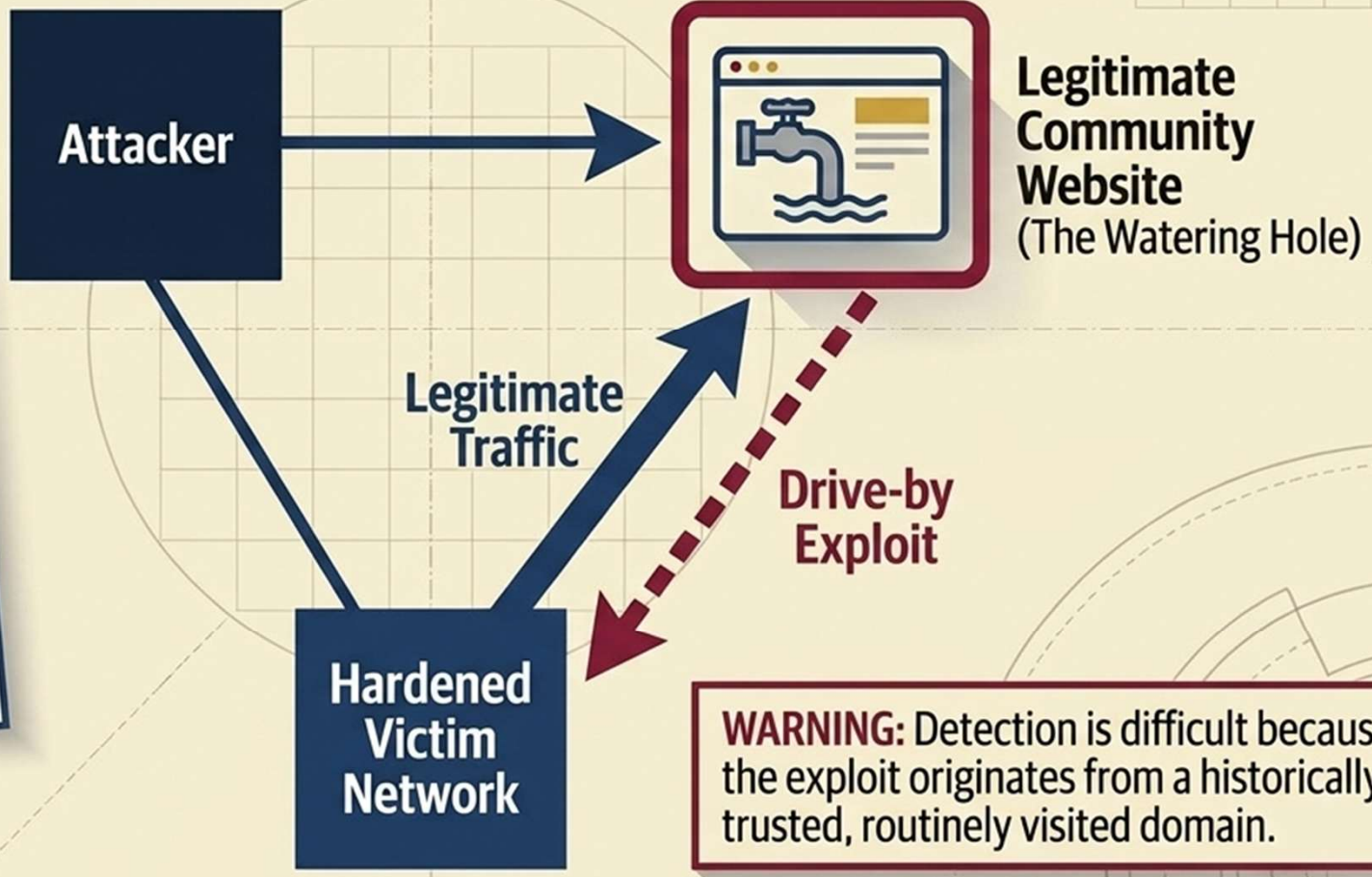
Delivery Phase Deep Dive: The Watering Hole Attack

Concept

- Infecting third-party websites that members of a specific target group are known to frequent.

Real-World Examples

- The VOHO affair; Hong Kong Telecom (2018) exploiting Adobe Flash vulnerability CVE-2018-4878.

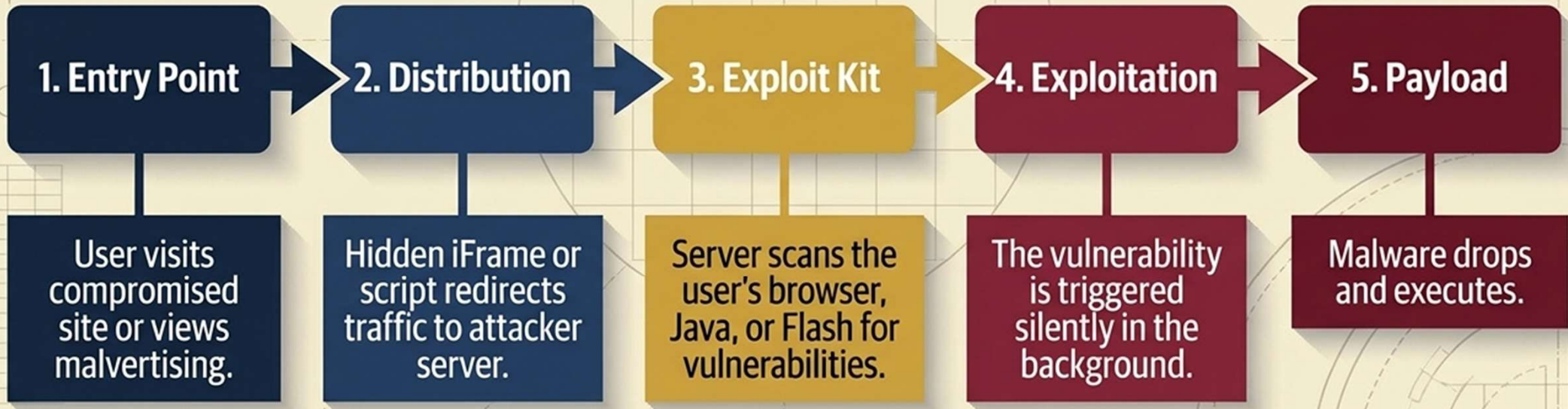


WARNING: Detection is difficult because the exploit originates from a historically trusted, routinely visited domain.

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Exploitation Phase: Anatomy of a Drive-by Download

ZERO-CLICK INFECTION: The user only needs to view the page.



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Synthesis: Initial Access Vector Matrix

Vector	Targeting Precision	Required User Interaction	Primary Defense
Spear-Phishing	High (Specific individual)	Yes (Must click/download)	DMARC, EDR, User Awareness
Waterholing	Medium (Specific demographic)	No (Just visit site)	Sandboxed browsers, Threat Intel
Drive-by Download	Low (Anyone visiting)	No (Just visit site)	Patch management, Exploit Kit blocking

Modern initial access relies heavily on bypassing perimeter defenses by piggybacking on authorized user outbound traffic.

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Installation Phase: Establishing a Foothold

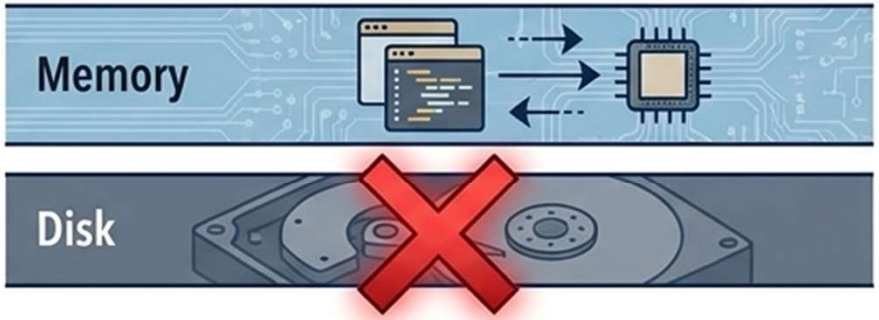
Precursor Malware & Droppers



Use of initial access brokers (e.g., QakBot, Emotet) to drop primary payloads.

Evasion Techniques

Living off the Land (LotL)



- Fileless execution operating entirely in memory.
- Abusing native tools like PowerShell or PsExec to disable Windows Defender.
- Avoiding disk writes to bypass traditional antivirus.
- LSASS memory dumping via tools like Mimikatz to extract credentials.

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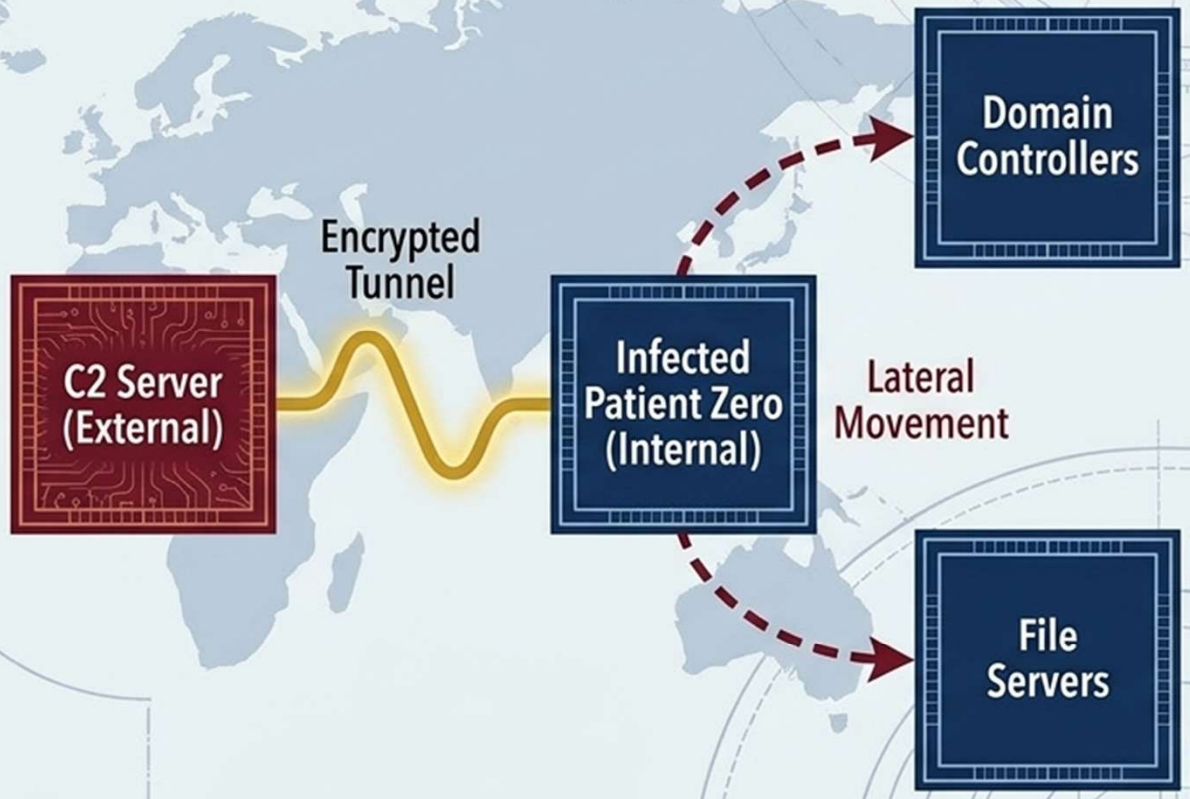
Command & Control (C2) and Post-Exploitation

Stealth Communications

Attackers hide C2 traffic using encrypted tunnels (HTTPS port 443) or DNS tunneling to blend in with legitimate outbound traffic.

Lateral Movement Tools

Heavy reliance on commercial penetration testing suites like **Cobalt Strike** (beacons) disguised as legitimate Windows processes.



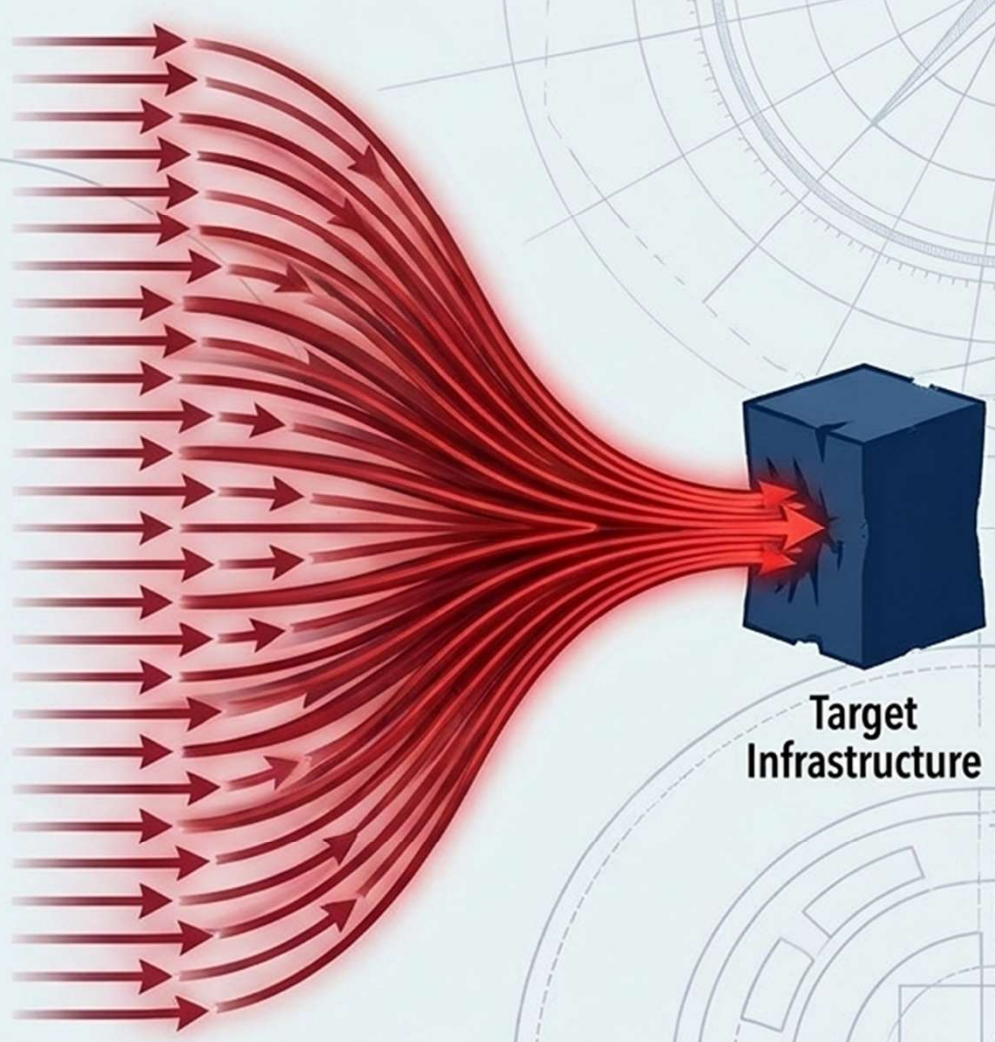
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Actions on Objectives: Sabotage & DDoS

Massive Volumetric Disruption

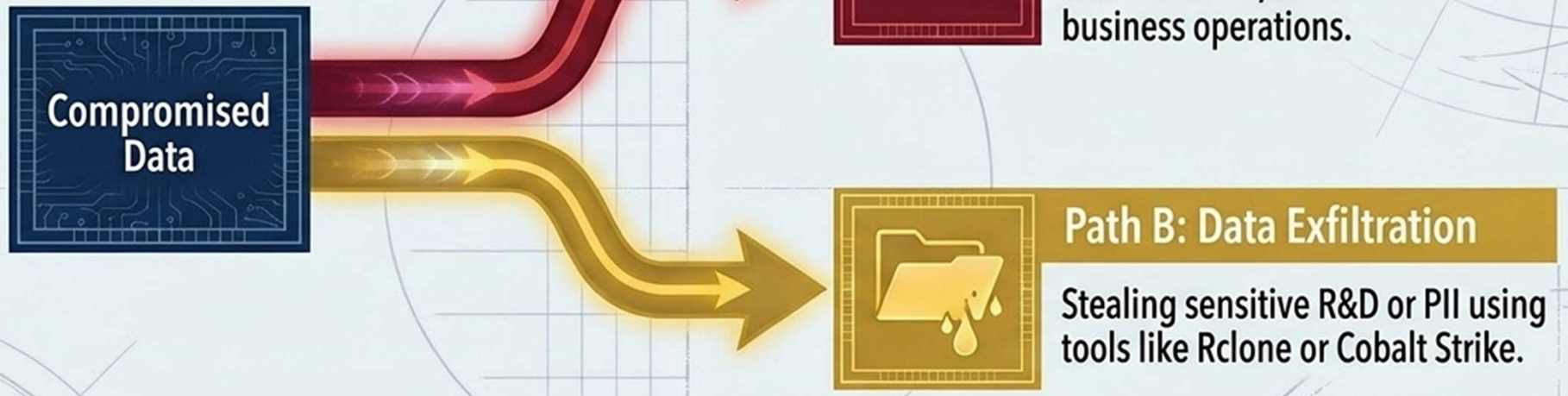
Case Study: The Mirai Botnet

Reconnaissance	Mass scanning for exposed IoT devices.
Access	Brute-forcing default Telnet and SSH credentials (admin/admin).
Action	Enslaving devices to launch massive TCP SYN, UDP, and HTTP GET floods.
Scale	Generating attacks exceeding 1 Tbps, effectively blacking out major DNS providers.



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Actions on Objectives: Double Extortion & Infostealers



The Double Extortion Paradigm

If the victim refuses to pay to decrypt their files, the attacker threatens to leak the stolen data on the Dark Web.

Example: The Colonial Pipeline attack (100GB exfiltrated, $5M ransom).

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The Next Evolution: AI in the Threat Landscape



Automated Reconnaissance

LLMs digesting massive OSINT datasets to find precise vulnerabilities instantly.

Deepfake Phishing

Highly personalized, grammatically perfect spear-phishing and voice vishing generated at scale.

Autonomous Agents

Moving beyond static scripts to AI agents that dynamically adapt their lateral movement based on network defenses.

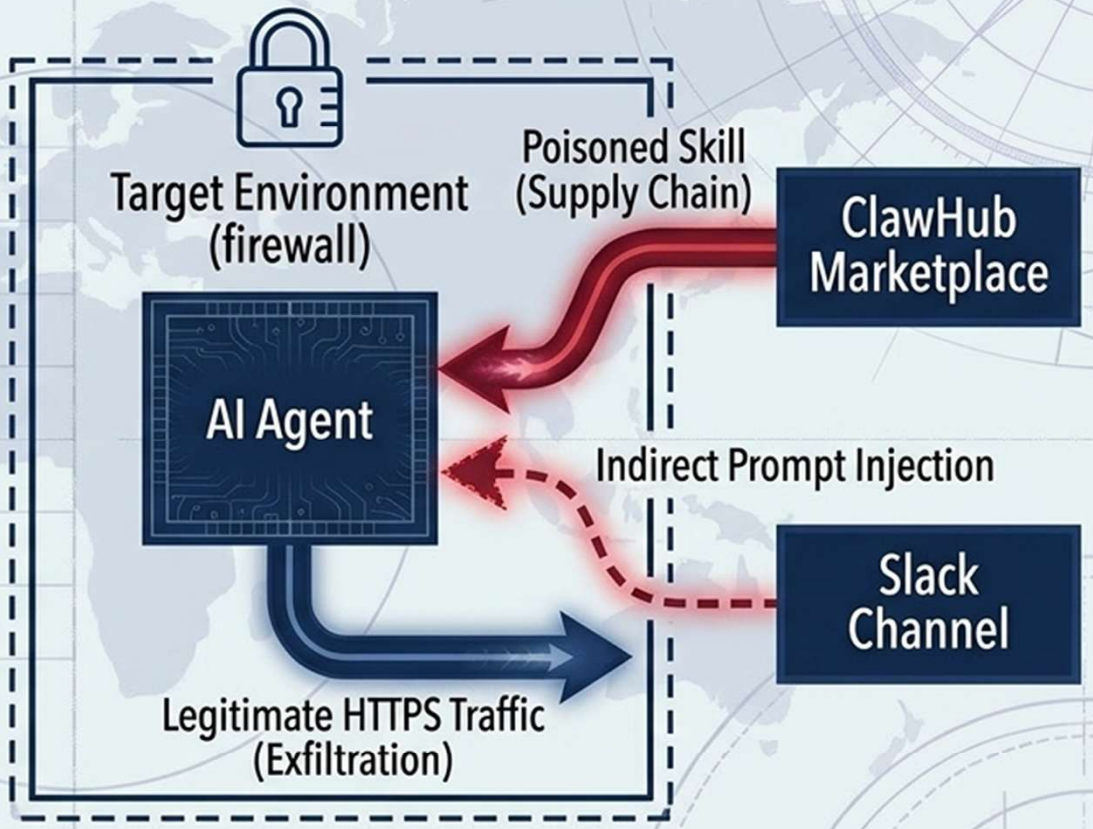
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Case Study: The OpenClaw AI Operation

Supply Chain:
Compromised via a poisoned skill on the ClawHub marketplace.

Execution:
The agent is hijacked via malicious text hidden in legitimate Slack messages.

Exfiltration:
The AI uses its own legitimate HTTPS terminal access to execute network commands and exfiltrate R&D data.



Invisible to WAF/EDR because the traffic appears as normal AI queries.

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Defensive Strategies: Breaking the Kill Chain



Delivery Shield

Protective DNS, Email Gateway Filtering, DMARC.

Exploitation Shield

Phishing-resistant MFA, Patch Management, Disabling unused plugins.

Installation Shield

Zero Trust Architecture (ZTA), EDR, disabling PowerShell for non-admins.

C2 & Actions Shield

Network Segmentation, Offline immutable backups (Cloud-to-Cloud).

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Incident Response & Conclusion

Detection

- Hunt for precursor malware (QakBot).
- Detect anomalous usage of native tools (PsExec, RDP).
- Review Sysmon logs.

Containment

- Isolate systems at the switch level.
- Do not power down immediately if memory forensics are needed to preserve artifacts.

Eradication

- Rebuild from known golden images.
- Reset all credentials.
- Implement Local Administrator Password Solution (LAPS).

As attack vectors evolve from manual drive-by exploits to autonomous AI agents, defense must shift from static perimeters to dynamic, identity-based Zero Trust architectures.

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